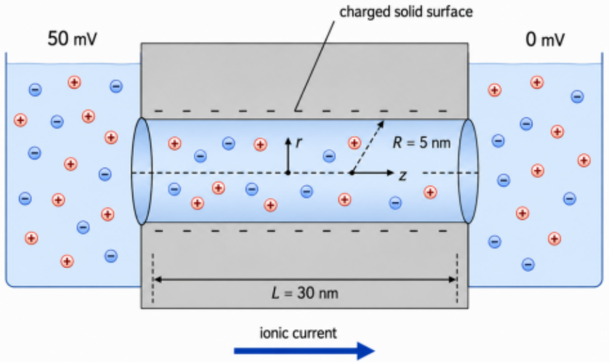
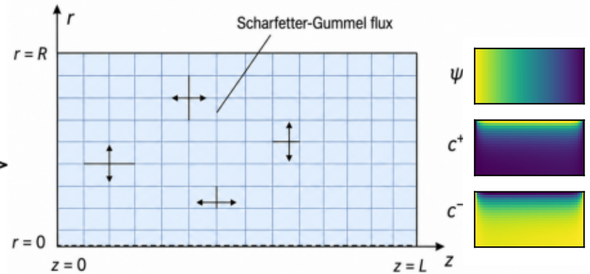
A. Charged nanopore model



discretize PNP

B. SG/FV reference solution



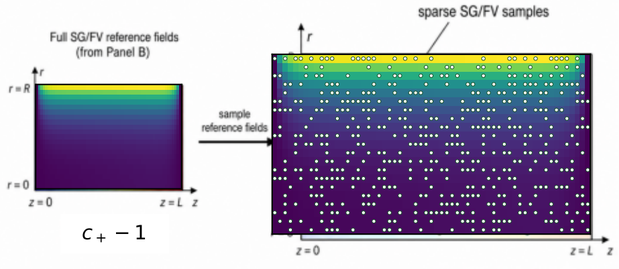
Equilibrium tests | Conservation

Grid and tolerance checks

Verified numerical reference

sample sparse anchors

C. Sparse numerical anchors

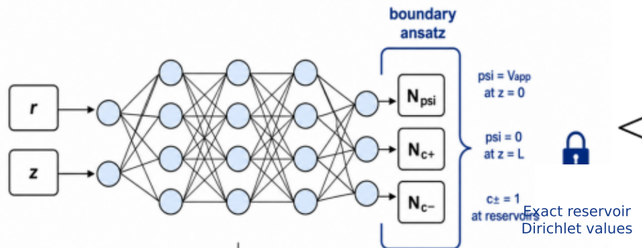


500 distinct anchors / 1281 nodes (39%)

Saved reference fields and sampling locations

train with physics + data

D. Exact-reservoir DA-PINN

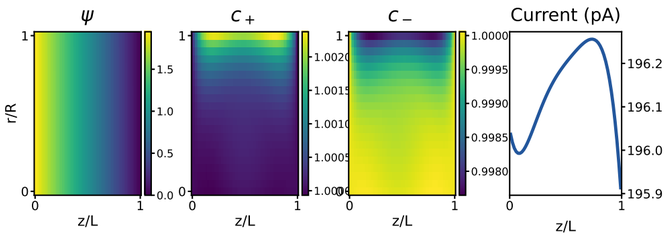


Four hidden layers, 64 SiLU units each

PNP + wall/axis + field/charge losses

predict fields and current

E. Fixed-state field prediction



1.366 ms median full-field evaluation

Timing excludes current calculation and training

Fixed operating state; no generalization claim